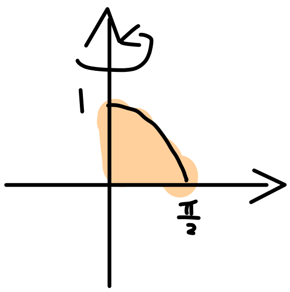
1. (Spring 2024, #4) Consider the region in the first quadrant that is bounded by $y = \cos x$, $x = 0$, and $y = 0$. A solid is formed by rotating this region about the y-axis.

   (a) Find the volume of this solid.



shell - method

$$V = \int_0^{\frac{\pi}{2}} 2\pi x \cdot \cos x \, dx.$$

$$= 2\pi \int_0^{\frac{\pi}{2}} x \, d\sin x$$

$$= 2\pi \left[x \sin x \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \, dx \right]$$

$$= 2\pi \left[\frac{\pi}{2} + \cos x \Big|_0^{\frac{\pi}{2}} \right].$$

$$= 2\pi \left[\frac{\pi}{2} - 1 \right].$$

   (b) Find a definite integral whose value is equal to the surface area of the curved surface of the solid (not including the circular flat base). You do not need to evaluate the integral.

$$S = \int 2\pi x \, ds.$$

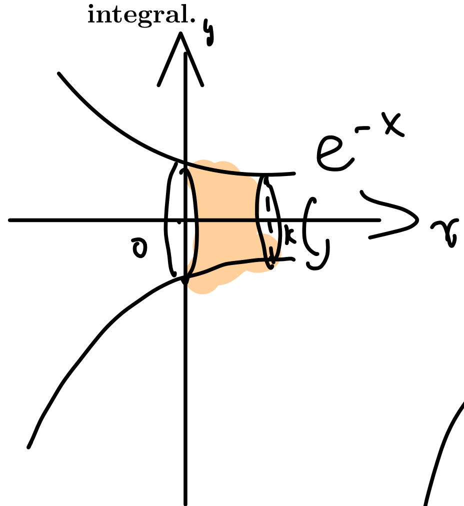
$$= \int_0^{\frac{\pi}{2}} 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

$$= \int_0^{\frac{\pi}{2}} 2\pi x \sqrt{1 + (-\sin x)^2} \, dx.$$

$$= \int_0^{\frac{\pi}{2}} 2\pi x \sqrt{1 + (\sin x)^2} \, dx.$$

2. (Fall 2023, #3) Consider the surface created by revolving the curve $y = e^{-x}$ about the x-axis where $0 \le x \le k$.

(a) Write down an integral expressing the area of this surface. **You do not need to evaluate this integral.**



$$S = \int 2\pi y \, ds.$$

$$= \int_{e^{-k}}^{e^{0}} 2\pi y \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy.$$

$y = e^{-x} \Rightarrow -x = \ln y \Rightarrow x = -\ln y.$

$\Rightarrow \frac{dx}{dy} = -\frac{1}{y}.$

$$= \int_{e^{-k}}^{1} 2\pi y \sqrt{1+\frac{1}{y^2}}\, dy.$$

(b) Let $k \to \infty$. Does the improper integral converge? Justify your answer.

$$\int_0^1 2\pi y \sqrt{1+\frac{1}{y^2}}\, dy.$$

$$= \int_0^1 2\pi \sqrt{y^2+1}\, dy.$$

$\forall\, x > 1 \Rightarrow \sqrt{x} \le x.$

$$\le \int_0^1 2\pi (y^2+1)\, dy$$

$$= \frac{2\pi}{3} y^3 \Big|_0^1 + 2\pi = \frac{2\pi}{3} + 2\pi < \infty.$$

$\Rightarrow$ Converge!

3. (Spring 2020, #6) Find the **length** of the curve $x = \frac{y^4}{8} + \frac{1}{4y^2}$ from $y = 1$ to $y = 2$.

$$L = \int_1^2 \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \,.$$

$$= \int_1^2 \sqrt{1 + \left[\frac{4y^3}{8} + \frac{1}{4}\,\frac{-2}{y^3}\right]^2}\, dy \,.$$

$$= \int_1^2 \sqrt{1 + \left(\frac{y^3}{2} - \frac{1}{2y^3}\right)^2}\, dy \,.$$

$$= \int_1^2 \sqrt{1 + \left(\frac{y^3}{2}\right)^2 + \left(\frac{1}{2y^3}\right)^2 - 2\cdot\frac{y^3}{2}\cdot\frac{1}{2y^3}}\, dy \,.$$

$$= \int_1^2 \sqrt{1 + \left(\frac{y^3}{2}\right)^2 + \left(\frac{1}{2y^3}\right)^2 - \frac{1}{2}}\, dy \,.$$

$$= \int_1^2 \sqrt{\Big(\underbrace{\frac{y^3}{2}}_{a}\Big)^2 + \Big(\underbrace{\frac{1}{2y^3}}_{b}\Big)^2 + \underbrace{\frac{1}{2}}_{2ab}}\; dy$$

$$2ab = 2\cdot\frac{y^3}{2}\cdot\frac{1}{2y^3} = \frac{1}{2},$$

$$= \int_1^2 \sqrt{\left(\frac{y^3}{2} + \frac{1}{2y^3}\right)^2}\, dy \,.$$

$$= \int_1^2 \left(\frac{y^3}{2} + \frac{1}{2y^3}\right) dy \,. \quad \text{(Note that } y>0\text{)}.$$

$$= \frac{y^4}{2\cdot 4} - \frac{1}{2}\cdot\frac{1}{2y^2}\Big|_1^2 = \left(\frac{2^4}{8} - \frac{1}{4\cdot 2^2}\right) - \left(\frac{1^4}{8} - \frac{1}{4\cdot 1^2}\right)$$

$$= \left(2 - \frac{1}{16}\right) - \left(\frac{1}{8} - \frac{1}{4}\right) = 2\tfrac{1}{16}\,.$$

4. (Fall 2019, #1b) Find an integral representing the length of the function $f(x) = \frac{x^3}{6} + \frac{1}{2x}$ on the interval $\left[\frac{1}{2}, 1\right]$.

$$\int_{\frac{1}{2}}^{1} \sqrt{1+(f'(x))^2}\, dx.$$

$$f'(x) = \frac{3x^2}{6} - \frac{1}{2x^2} = \frac{x^2}{2} - \frac{1}{2x^2}$$

$$= \int_{\frac{1}{2}}^{1} \sqrt{1+\left(\frac{x^2}{2} - \frac{1}{2x^2}\right)^2}\, dx.$$

$$= \int_{\frac{1}{2}}^{1} \sqrt{1+\left(\frac{x^2}{2}\right)^2 - 2\cdot\frac{x^2}{2}\cdot\frac{1}{2x^2} + \left(\frac{1}{2x^2}\right)^2}\, dx$$

$$= \int_{\frac{1}{2}}^{1} \sqrt{1+\left(\frac{x^2}{2}\right)^2 - \frac{1}{2} + \left(\frac{1}{2x^2}\right)^2}\, dx$$

$$= \int_{\frac{1}{2}}^{1} \sqrt{\underbrace{\left(\frac{x^2}{2}\right)^2}_{a} + \underbrace{\frac{1}{2}}_{2ab.} + \underbrace{\left(\frac{1}{2x^2}\right)^2}_{b}}$$

$2ab = 2\cdot\frac{x^2}{2}\cdot\frac{1}{2x^2} = \frac{1}{2}$.

$$= \int_{\frac{1}{2}}^{1} \sqrt{\left(\frac{x^2}{2} + \frac{1}{2x^2}\right)^2}\, dx.$$

$$= \int_{\frac{1}{2}}^{1} \left(\frac{x^2}{2} + \frac{1}{2x^2}\right) dx.$$

$$= \frac{x^3}{6} - \frac{1}{2x}\Big|_{\frac{1}{2}}^{1} = \left(\frac{1}{6} - \frac{1}{2}\right) - \left(\frac{\left(\frac{1}{2}\right)^3}{6} - \frac{1}{2\cdot\frac{1}{2}}\right) = \frac{31}{48}.$$

5. (Fall 2019, #2) Consider the surface S obtained by rotating the curve $y = e^{-x}$ about the x-axis, with $0 \le x < \infty$.

(a) Establish that the **surface area** of S is given by $A = 2\pi \int_0^1 \sqrt{\cancel{a} + u^2}\,du$.
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$$A = \int 2\pi y \, ds$$

$x = -\ln y$.

$$= \int_0^1 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy .$$

$$= \int_0^1 2\pi y \sqrt{1 + \left(\frac{1}{y}\right)^2} \, dy .$$

$$= \int_0^1 2\pi \sqrt{y^2 + 1} \, dy .$$

(b) Calculate the value of the integral A.

$$\int_0^1 2\pi \sqrt{y^2+1} \, dy .$$

$$\overset{y = \tan\theta}{=} \int_0^{\frac{\pi}{4}} 2\pi \sqrt{(\tan\theta)^2 + 1} \, (\sec\theta)^2 \, d\theta .$$

$$= \int_0^{\frac{\pi}{4}} 2\pi \cdot (\sec\theta)(\sec\theta)^2 \, d\theta .$$

$$= \int_0^{\frac{\pi}{4}} 2\pi \sec\theta \, d\tan\theta .$$

$(\sec\theta)' = \sec\theta \tan\theta$.

$$= 2\pi \sec\theta \tan\theta \Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 2\pi (\sec\theta \tan\theta) \tan\theta \, d\theta .$$

$$= 2\pi \cdot \sqrt{2} . - \int_0^{\frac{\pi}{4}} 2\pi \sec\theta (\sec^2\theta - 1) \, d\theta .$$

$\tan^2\theta = \sec^2\theta - 1$.

$$= 2\pi\sqrt{2} - \int_0^{\frac{\pi}{4}} 2\pi (\sec\theta)^3 d\theta + \int_0^{\frac{\pi}{4}} 2\pi \sec\theta \, d\theta .$$

$$\int_0^{\frac{\pi}{4}} 2\pi (\sec\theta)^3 d\theta .$$

$$= \frac{1}{2}\left(2\pi\sqrt{2} + 2\pi \int_0^{\frac{\pi}{4}} \sec\theta \, d\theta\right],$$

$$= \pi\left(\sqrt{2} + \ln(\sec\theta + \tan\theta)\Big|_0^{\frac{\pi}{4}}\right]$$

$$= \pi\Big(\sqrt{2} + \ln(\underbrace{\sec\tfrac{\pi}{4}}_{=\sqrt{2}} + \underbrace{\tan\tfrac{\pi}{4}}_{=1}) - \underbrace{\ln(\sec 0 + \tan 0)}_{= \ln(1) = 0}\Big).$$

$$= \pi(\sqrt{2} + \ln(\sqrt{2} + 1)).$$